

SHETH L.U.J AND SIR M.VCOLLEGE

Subject: Data Analysis with SAS / SPSS /R

Practical no. 15

Aim: Generating basic summaries
using str() or summary() (R).

The screenshot displays the RStudio interface with the following components:

- Source Panel:** Contains R code for loading a CSV file and generating summaries.

```
> df <- read.csv("retail_store_sales.csv")
> print("--- DATA LOADED ---")
[1] "--- DATA LOADED ---"
> head(df)
  Transaction.ID Customer.ID Category Item Price.Per.Unit
1 TXN_6867343    CUST_09 Patisserie Item_10_PAT    18.5
2 TXN_3731986    CUST_22 Milk Products Item_17_MILK    29.0
3 TXN_9303719    CUST_02 Butchers Item_12_BUT    21.5
4 TXN_9458126    CUST_06 Beverages Item_16_BEV    27.5
5 TXN_4575373    CUST_05 Food Item_6_FOOD    12.5
6 TXN_7482416    CUST_09 Patisserie NA
  Quantity Total.Spent Payment.Method Location Transaction.Date
1      10      185.0 Digital wallet Online 2024-04-08
2       9      261.0 Digital wallet Online 2023-07-23
3       2       43.0 Credit Card Online 2022-10-05
4       9      247.5 Credit Card Online 2022-05-07
5       7       87.5 Digital wallet Online 2022-10-02
6      10      200.0 Credit Card Online 2023-11-30
  Discount.Applied
1      True
2      True
3     False
4      True
5     False
6     False
> # -----
> # STRUCTURE OF THE DATA
> # -----
> print("--- OUTPUT OF str() ---")
[1] "--- OUTPUT OF str() ---"
> str(df)
'data.frame': 12575 obs. of 11 variables:
 $ Transaction.ID : chr "TXN_6867343" "TXN_3731986" "TXN_9303719" "TXN_9458126" ...
 $ Customer.ID : chr "CUST_09" "CUST_22" "CUST_02" "CUST_06" ...
 $ Category : chr "Patisserie" "Milk Products" "Butchers" "Beverages" ...
 $ Item : chr "Item_10_PAT" "Item_17_MILK" "Item_12_BUT" "Item_16_BEV" ...
 $ Price.Per.Unit : num 18.5 29 21.5 27.5 12.5 NA 5 33.5 27.5 36.5 ...
```
- Environment Panel:** Lists loaded objects: tidy_sales (1000 obs. of 15 variables), unique_customers (25 obs. of 11 variables), volatile_days (114 obs. of 7 variables), and wide_df (12575 obs. of 6 variables).
- Files Panel:** Shows the file explorer with 'retail_store_sales.csv' selected.

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```
R - R4.5.2 - ~/
5 False
6
> # -----
> # STRUCTURE OF THE DATA
> # -----
> print("--- OUTPUT OF str() ---")
[1] "--- OUTPUT OF str() ---"
> str(df)
'data.frame': 12575 obs. of 11 variables:
 $ Transaction.ID : chr "TXN_6867343" "TXN_3731986" "TXN_9303719" "TXN_9458126"
 ...
 $ Customer.ID : chr "CUST_09" "CUST_22" "CUST_02" "CUST_06" ...
 $ Category : chr "Patisserie" "Milk Products" "Butchers" "Beverages" ...
 $ Item : chr "Item_10_PAT" "Item_17_MILK" "Item_12_BUT" "Item_16_BE"
 ...
 $ Price.Per.Unit : num 18.5 29 21.5 27.5 12.5 NA 5 33.5 27.5 36.5 ...
 $ Quantity : num 10 9 2 9 7 10 8 NA 1 3 ...
 $ Total.Spent : num 185 261 43 247.5 87.5 ...
 $ Payment.Method : chr "Digital wallet" "Digital wallet" "Credit Card" "Credit Card" ...
 $ Location : chr "Online" "Online" "Online" "Online" ...
 $ Transaction.Date : chr "2024-04-08" "2023-07-23" "2022-10-05" "2022-05-07" ...
 $ Discount.Applied : chr "True" "True" "False" "" ...
> # -----
> # STATISTICAL SUMMARY
> # -----
> print("--- OUTPUT OF summary() ---")
[1] "--- OUTPUT OF summary() ---"
> summary(df)
Transaction.ID Customer.ID Category Item
Length:12575 Length:12575 Length:12575 Length:12575
Class :character Class :character Class :character Class :character
Mode :character Mode :character Mode :character Mode :character

Price.Per.Unit Quantity Total.Spent Payment.Method
Min. : 5.00 Min. : 1.000 Min. : 5.0 Length:12575
1st Qu.: 14.00 1st Qu.: 3.000 1st Qu.: 51.0 Class :character
Median : 23.00 Median : 6.000 Median : 108.5 Mode :character
Mean : 23.37 Mean : 5.536 Mean : 129.7
3rd Qu.: 33.50 3rd Qu.: 8.000 3rd Qu.: 192.0
Max. : 41.00 Max. : 10.000 Max. : 410.0
NA's : 609 NA's : 604 NA's : 604
```

Environment History Connections Tutorial

R - Global Environment

- tidy_sales 1000 obs. of 15 variables
- unique_customers 25 obs. of 11 variables
- volatile_days 114 obs. of 7 variables
- wide_df 12575 obs. of 6 variables

Values

avg_dist	6.62475
avg_spent	129.652577061231
avg_time	44.7445
max_price	41

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Name	Size	Modified
desktop.ini	402 B	Dec 8, 2024, 2:50 PM
Downloads - Shortcut.lnk	737 B	Mar 18, 2025, 3:29 PM
flower_dataset - flower_dataset.csv	283.7 KB	Dec 8, 2025, 11:03 AM
Iris - Iris.csv	5 KB	Dec 8, 2025, 11:06 AM
My Music		
My Pictures		
My Videos		
NetBeansProjects		
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pushpal prac		
retail_store_sales.csv	1.1 MB	Dec 8, 2025, 1:25 PM
sales_data.csv	101.2 KB	Dec 8, 2025, 12:04 PM
sales_dataset.csv	698 B	Dec 2, 2025, 12:18 PM
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RStudio

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Source

```
R - R4.5.2 - ~/
> # -----
> # IMPROVE SUMMARY BY CONVERTING CATEGORY TO FACTOR
> # -----
> df$category <- as.factor(df$category)
> print("--- SUMMARY AFTER FACTOR CONVERSION ---")
[1] "--- SUMMARY AFTER FACTOR CONVERSION ---"
> summary(df)
Transaction.ID Customer.ID Category
Length:12575 Length:12575 Electric household essentials:1591
Class :character Class :character Furniture :1591
Mode :character Mode :character Food :1588
Milk Products :1584
Butchers :1568
Beverages :1567
(Other) :3086

Item Price.Per.Unit Quantity Total.Spent
Length:12575 Min. : 5.00 Min. : 1.000 Min. : 5.0
Class :character 1st Qu.: 14.00 1st Qu.: 3.000 1st Qu.: 51.0
Mode :character Median : 23.00 Median : 6.000 Median : 108.5
Mean : 23.37 Mean : 5.536 Mean : 129.7
3rd Qu.: 33.50 3rd Qu.: 8.000 3rd Qu.: 192.0
Max. : 41.00 Max. : 10.000 Max. : 410.0
NA's : 609 NA's : 604 NA's : 604
```

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R - Global Environment

- tidy_sales 1000 obs. of 15 variables
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- volatile_days 114 obs. of 7 variables
- wide_df 12575 obs. of 6 variables

Values

avg_dist	6.62475
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Source

```

R - R4.5.2 ~ /
> df$category <- as.factor(df$category)
> print("--- SUMMARY AFTER FACTOR CONVERSION ---")
[1] "--- SUMMARY AFTER FACTOR CONVERSION ---"
> summary(df)
Transaction.ID      Customer.ID      Category
Length:12575      Length:12575      Electric household essentials:1591
Class :character    Class :character    Furniture :1591
Mode :character      Mode :character      Food :1588
                        Milk Products :1584
                        Butchers :1568
                        Beverages :1567
                        (Other) :3086

Item      Price.Per.Unit      Quantity      Total.Spent
Length:12575      Min. : 5.00      Min. : 1.000      Min. : 5.0
Class :character    1st Qu.:14.00    1st Qu.: 3.000    1st Qu.: 51.0
Mode :character      Median :23.00    Median : 6.000    Median :108.5
                        Mean :23.37      Mean : 5.536      Mean :129.7
                        3rd Qu.:33.50    3rd Qu.: 8.000    3rd Qu.:192.0
                        Max. :41.00      Max. :10.000      Max. :410.0
                        NA's :609      NA's :604      NA's :604

Payment.Method      Location      Transaction.Date      Discount.Applied
Length:12575      Length:12575      Length:12575      Length:12575
Class :character    Class :character    Class :character    Class :character
Mode :character      Mode :character      Mode :character      Mode :character

> # -----
> # SPECIFIC SUMMARY VALUES
> # -----
> avg_spent <- mean(df$Total.Spent, na.rm = TRUE)
> max_price <- max(df$Price.Per.Unit, na.rm = TRUE)
> print(paste("Average Total Spent:", avg_spent))
[1] "Average Total Spent: 129.652577061231"
> print(paste("Highest Price Per Unit:", max_price))
[1] "Highest Price Per Unit: 41"
>

```

Environment History Connections Tutorial

R - Global Environment

Object	Size
tidy_sales	1000 obs. of 15 variables
unique_customers	25 obs. of 11 variables
volatile_days	114 obs. of 7 variables
wide_df	12575 obs. of 6 variables

Values

Variable	Value
avg_dist	6.62475
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